

Week7_individual_assignment_Sulaiman (Stan) Abu Romman

Part 3 – Individual Research Task: Sequence Diagram

A Sequence Diagram is a UML interaction diagram that shows how objects communicate with each other over time. It focuses on the order in which messages are sent between objects during a specific scenario. Unlike a System Sequence Diagram, which treats the system as a black box, a sequence diagram shows the internal objects of the system and how they collaborate to perform a task.

The main components of a sequence diagram are objects or actors, lifelines, messages, and activation bars. Objects are placed horizontally at the top of the diagram, while time flows vertically from top to bottom. Messages between objects are represented by arrows, showing the direction and sequence of communication.

The purpose of a sequence diagram in software development is to support the design phase by explaining how the system works internally. It helps developers understand object responsibilities, message flow, and interaction logic before implementation. Sequence diagrams are especially useful for translating use cases into detailed system behavior.

Application – Sequence Diagram for Stomach Saver

The following sequence diagram represents the internal process of placing a food order in the Stomach Saver system. It shows how the customer interacts with system objects and how these objects collaborate to complete the order and payment process.

